

AlignRuScore: Adapting AlignScore to Russian Language

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ABSTRACT

Ensuring factual consistency in generated text is crucial for reliable natural language processing applications. We introduce **ALIGNRuSCORE**, a comprehensive adaptation of the AlignScore metric for Russian. This unified evaluation covers natural language inference, fact verification, paraphrase detection, semantic textual similarity, question answering, and information retrieval. We compiled and translated over 118,000 examples—combining major English benchmarks with Russian-native datasets such as RuFacts and RuSTS—and fine-tuned a RuBERT-based alignment model with task-specific classification and regression heads. On held-out test sets, **ALIGNRuSCORE** achieves $F_1 \geq 0.81$ on controlled entailment and fact verification (SNLI, FEVER), $F_1 \approx 0.77$ and $AUC \approx 0.88$ on paraphrase detection (QQP), and $R^2 \approx 0.86$ on English-derived STS (SICK). Performance is lower in open-domain QA ($F_1 \approx 0.29$ on MS MARCO) and Russian STS ($R^2 \approx 0.56$), highlighting avenues for future work. We release our translated corpora, model checkpoints, and code to support further research. Our results demonstrate that a unified alignment metric can be successfully ported to Russian, laying the groundwork for robust multilingual factual consistency evaluation.

1 INTRODUCTION

Modern natural language generation systems are increasingly used in critical applications where factual correctness is imperative. AlignScore [3] introduced a unified alignment function that maps pairs of text—namely, a source (context) and a target (claim)—to a factual consistency score. While originally designed and validated on English corpora, applying AlignScore directly to Russian texts may underperform due to specific linguistic characteristics such as complex morphology, inflection, and syntactic flexibility. This paper proposes an adaptation strategy that re-engineers the AlignScore metric to improve its reliability when evaluating factual consistency in Russian generated content.

2 RELATED WORK

AlignScore has emerged as a powerful metric for measuring factual consistency by unifying diverse natural language processing tasks, including natural language inference (NLI), question answering (QA), fact verification, paraphrase, and Semantic Textual Similarity. Other factual consistency metrics—such as QAFactEval and UniEval—have shown promising results but remain primarily tuned on English data. Our work is to fine-tune the methodology behind AlignScore for the Russian context.

There are two native fact alignment datasets that we found. RuSTS is human translation of STSB. RuFact [1] is a paraphrase dataset created with various approaches for generation of training examples.

3 METHODOLOGY

Our methodology adapts the original AlignScore framework [3] to the Russian language, resulting in AlignRuScore. The process involved data collection and translation, unified model training, adaptation of the metric’s calculation strategy and evaluation of resulting metric.

3.1 Data Collection and Translation

We constructed a diverse Russian training corpus by translating subsets (up to 10,000 examples each where applicable) of the English datasets used in the original AlignScore paper, covering tasks such as Natural Language Inference (NLI), Fact Verification, Paraphrasing, Question Answering (QA), and Semantic Textual Similarity (STS). Machine translation was performed using Yandex Translate. This translated data was supplemented with Russian-native datasets, specifically RuFacts [1] for paraphrase/fact verification and the RuSTS Benchmark for STS. The final unified corpus comprised over 118,000 training examples spanning multiple alignment-related tasks as shown in Table 1.

NLP Task	Dataset	Training Task	Size
NLI	SNLI	3-way classification	10k
	MultiNLI	3-way classification	10k
	Adversarial NLI	3-way classification	10k
	DocNLI	binary classification	10k
Fact Verification	NLI-style FEVER	3-way classification	10k
	Vitamin C	3-way classification	10k
Paraphrase	QQP	binary classification	10k
	PAWS	binary classification	10k
	RuFacts*	binary classification	6.7k
STS	SICK	regression	4.4k
	Ru STS Benchmark*	regression	7.8k
QA	RACE	binary classification	10k
Information Retrieval	MS MARCO	binary classification	10k
Overall	All datasets	all tasks	118.9k

Table 1: The training datasets of our alignment model. Datasets marked with a * were already translated or created in Russian. The other datasets were translated as a part of our work.

3.2 Adapting AlignScore Metric to Russian

Training Unified Alignment Function on Russian Corpus.

Following [3], we employed a RuBERT-base model [2] as the backbone for our alignment function. The model was fine-tuned using a unified multi-task learning approach on the compiled Russian corpus. It features task-specific output heads for:

- 3-way classification (predicting entailment, neutral, or contradiction)
- Binary classification (predicting aligned or not-aligned)
- Regression (predicting a similarity score between 0 and 1)

The model was trained by optimizing a combined loss function encompassing cross-entropy for classification heads and mean squared error for the regression head, allowing it to learn a generalized representation of textual alignment from the diverse data sources. Training hyperparameters are detailed in Table 2.

Hyperparameter	AlignRuScore-base
Base Model	RuBERT-base
Parameters	180M
Batch Size	12
Epochs	3
Optimizer	AdamW
Learning Rate	1e-5
Weight Decay	0.1
Adam ϵ	1e-6
Warmup Ratio	0.06
Random Seed	2025
GPU	1/4 A100
GPU Hour	4h

Table 2: The hyperparameters used to train the alignment model.

Following AlignScore’s original formulation, we train a small feed-forward network for each type of training task 1 and learn them simultaneously, so our embedding space and heads can properly handle alignment for all nlp alignment tasks.

Adapting AlignScore metric.

To evaluate the factual consistency between a given Russian context and claim, AlignRuScore utilizes the trained alignment function combined with the context/claim splitting and aggregation strategy proposed in [3]. Specifically:

- (1) The potentially long context is segmented into overlapping chunks of approximately 350 tokens at sentence boundaries.
- (2) The claim is split into individual sentences.
- (3) Each claim sentence is evaluated against every context chunk using the alignment model’s 3-way classification head. The probability assigned to the ‘aligned’ (entailment) class is used as the alignment score.
- (4) For each claim sentence, the maximum alignment score across all context chunks is selected, representing the strongest support found in the context for that specific sentence.
- (5) The final AlignRuScore is the average of these maximum scores across all sentences in the claim.

This chunking and aggregation approach allows the metric to handle long contexts effectively while producing a fine-grained

score sensitive to inconsistencies at the sentence level within the claim.

3.3 Evaluation on Russian datasets

We evaluated AlignRuScore on held-out test portions of Russian datasets corresponding to the core alignment tasks on which we trained metric. Performance was measured using standard metrics appropriate for each task: Accuracy, micro-F1, Precision, Recall, F1, and ROC AUC for classifications; and R^2 and MSE for regression.

4 RESULTS

4.1 Resulting Metric Evaluation

Evaluation results presented in Tables 3-5 reporting results across different types of tasks and metrics.

Table 3: Results on 3-way classification

Dataset	Precision	Recall	micro F1	Accuracy
SNLI	0.743	0.743	0.743	0.743
MultiNLI	0.674	0.674	0.674	0.674
ANLI	0.692	0.692	0.692	0.692
FEVER	0.814	0.814	0.814	0.814
VitaminC	0.596	0.596	0.596	0.596

Table 4: Results on binary classification

Dataset	Precision	Recall	F1	ROC AUC
RUFact	0.591	0.908	0.716	0.732
QQP	0.707	0.841	0.768	0.879
PAWS	0.613	0.896	0.728	0.766
DocNLI	0.762	0.954	0.847	0.897
RACE	0.250	0.998	0.400	0.524
MS MARCO	0.201	0.520	0.289	0.703

Table 5: Results on regression

Dataset	MSE	R^2
SICK	0.228	0.857
RU_STS	0.954	0.557

Across the first group of tasks (Table 3), the model achieves its strongest results on SNLI and FEVER; the model achieves over 0.81 F_1 and accuracy, indicating reliable entailment detection when sentence pairs are relatively controlled. By contrast, performance drops on MultiNLI and ANLI ($F_1 \approx 0.67$ – 0.69 , accuracy ≈ 0.67 – 0.69), reflecting challenges with genre diversity and adversarial examples. VitaminC fact-checking likewise yields a lower F_1 of 0.596 and accuracy of 0.596, suggesting that fine-grained claim verification under terse or noisy contexts remains an open issue.

In paraphrase detection, QQP achieves the highest balance ($F_1 = 0.768$, ROC AUC = 0.879), showing robust behavior on well-structured

question pairs. RUFact and PAWS, which contain noisier or adversarial paraphrase pairs, score lower ($F_1 \approx 0.72$, $AUC \approx 0.73\text{--}0.77$), indicating sensitivity to domain shifts. Document-level entailment (DocNLI) yields $F_1 = 0.847$ and $AUC = 0.897$, confirming strong binary entailment classification at scale. However, QA performance lags: on RACE, recall is near perfect (0.998) but precision falls to 0.250 (giving $F_1 = 0.400$), revealing a tendency to over-generate candidate answers. MS MARCO shows improved precision (0.201) and $F_1 = 0.289$, yet remains below paraphrase and inference levels, highlighting the need for more precise answer extraction.

Semantic similarity results are strong on SICK ($MSE = 0.228$, $R^2 = 0.857$), demonstrating reliable sentence embedding and regression in English. Performance degrades on RU-STS ($MSE = 0.954$, $R^2 = 0.557$); however, this degradation is not so significant.

4.2 Evaluating LLMs on our metric

To further validate the applicability of AlignRuScore, we conducted an evaluation of the Gemini 1.5-Flash model on a held-out subset of 200 examples. The model’s outputs were scored using our adapted metric, yielding a mean factual-consistency score of 0.7285 ± 0.0639 .

These results demonstrate that AlignRuScore provides a robust, unified evaluation framework for assessing factual consistency in Russian-language LLM outputs across a variety of task types.

5 CONCLUSION

In this work, we presented ALIGNRUSCORE, a Russian-language adaptation of the unified factual consistency metric AlignScore. By translating and augmenting a diverse set of English benchmarks—spanning natural language inference, fact verification, paraphrase detection, semantic textual similarity, question answering, and information retrieval—and fine-tuning a RuBERT-based alignment model, we obtained a single alignment function capable of handling the nuances of Russian morphology and syntax.

Our empirical evaluation on held-out test sets demonstrates that ALIGNRUSCORE achieves strong performance on entailment and fact verification tasks (e.g., $F_1 \geq 0.81$ on SNLI and FEVER) and maintains robust paraphrase detection (e.g., $F_1 \approx 0.77$ on QQP). At the same time, we observed areas for improvement in QA and IR settings (e.g., $F_1 \approx 0.29$ on MS MARCO) and in the STS domain (e.g., $R^2 \approx 0.56$ on RU-STS). These results highlight both the promise of a unified metric for Russian and the opportunity to further tailor the model to task-specific challenges.

Looking ahead, we plan to (1) incorporate additional Russian-native datasets—particularly for summarization and dialogue consistency; (2) explore architecture variants, such as multilingual transformer backbones and task-adaptive adapters; and (3) evaluate ALIGNRUSCORE in downstream applications, including automated fact-checking and evaluation of Russian-language generative models. We release our code, translated corpora, and trained checkpoints to facilitate further research and practical deployment in Russian NLP.

Overall, ALIGNRUSCORE represents a significant step toward reliable, language-agnostic evaluation of factual consistency, enabling more rigorous assessment of Russian text generation systems and laying the groundwork for broader multilingual alignment metrics.

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